INMO(21^{st}) – 2006

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. In a non equilateral triangle ABC, the sides a, b, c forms an arithmetic progression. Let I and O denote the incentre and circumcentre of the triangle respectively.
   (a) Prove that IO is perpendicular to BI.
   (b) Suppose BI extended meets AC in K, and D, E are the midpoints of BC, BA respectively. Prove that I is the circumcentre of triangle DKE.
2. Prove that for every positive integer n there exists a unique ordered pair (a, b) of positive integers such that $n = \frac{1}{2}(a+b-1)(a+b-2)+a$.
3. Let X denote the set of all triple (a, b, c) of integers. Define a function $f : X \to X$ by $f(a,b,c) = (a+b+c, ab+bc+ca, abc)$. Find all triples (a, b, c) in X such that $f(f(a,b,c)) = (a,b,c)$.
4. Some 46 squares are randomly selected from a 9X9 chess board and are coloured red. Show that there exist a 2X2 block of 4 squares of which at least there are coloured red.
5. In a cyclic quadrilateral ABCD, AB = a, BC = b, CD = c, $\angle ABC = 120^0, \angle ABD = 30^0$. Prove that
   (a) $c \geq a+b$
   (b) $\left|\sqrt{c+a} - \sqrt{c+b}\right| = \sqrt{c-a-b}$.
6.
   (a) Prove that if n is a positive integer such that $n \geq 4011^2$, then there exist an integer l such that $n < l^2 < \left(1+\frac{1}{2005}\right)n$.
   (b) Find the smallest positive integer M for which whenever an integer n is such that $n \geq M$, there exist an integer l such that $n < l^2 < \left(1+\frac{1}{2005}\right)n$.

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